



Analysis of Agricultural Data

Presented

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for the Degree of Bachelor of Statistics

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Abstract

This project investigates the application of multiple linear regression analysis to explore the relationships between various agricultural factors and crop production. Focusing on key variables such as rainfall, area cultivated, fertilizer usage, and pesticide application, the study aims to quantify their influence on crop yield. Through detailed analyses, the research provides insights into how these factors interact within different contexts, specifically examining season-wise, state-wise, and crop-wise variations in agricultural productivity.

The analysis begins with a thorough investigation of the data to understand trends and correlations among the variables. This includes visual representations that highlight the relationships and distributions within the dataset. The primary objective is to construct a robust multiple linear regression model that accurately reflects the dynamics of crop production based on the selected predictors.

In addition to the regression analysis, this project emphasizes the importance of diagnostic evaluations, including residual plots, to assess the validity of the model assumptions. By analyzing the residuals, we can identify potential areas for improvement and ensure that the model provides reliable predictions.

The findings from this study not only enhance our understanding of the factors influencing crop yield but also offer practical implications for stakeholders in the agricultural sector. By leveraging data-driven insights, farmers and policymakers can make informed decisions that contribute to improved agricultural practices and optimized resource allocation.

Ultimately, this project highlights the significance of multiple linear regression as a valuable tool for analyzing complex relationships in agricultural data, demonstrating its potential to support effective strategies for enhancing crop productivity and sustainability.

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Chapter 1

Introduction

Agriculture plays a vital role in the Indian economy, contributing significantly to employment and the Gross Domestic Product (GDP). With over half of India's population dependent on agriculture for their livelihood, understanding the factors that influence agricultural productivity is essential for sustainable development. Statistical analysis of agricultural data provides valuable insights into the patterns and trends that shape agricultural performance and help policymakers, researchers, and farmers make informed decisions.

This project focuses on the comprehensive analysis of agricultural data in India, covering multiple variables over time, seasons, states, and crop types. By examining factors such as rainfall, crop area, production, fertilizer and pesticide usage, and yield, the project aims to highlight the relationships and trends that can influence agricultural outcomes.

The analysis spans historical data across various regions and time periods, allowing for a deeper understanding of how agricultural practices and outcomes vary with geographic and climatic conditions, as well as temporal factors. This temporal and spatial diversity in the data is crucial for capturing seasonal cycles, state-wise variations, and crop-specific patterns that are unique to India's agricultural landscape.

The relevance of this analysis extends to addressing critical challenges such as optimizing crop yield, improving resource management, and adapting to climate change. As India continues to advance its agricultural sector, statistical analyses such as this one provide a scientific foundation for strategic planning and policy formulation aimed at achieving long-term agricultural sustainability.

1.1 Looking at the data

Crop	Crop_Year	Season	State	Area	Production	Annual_Rainfall	Fertilizer	Pesticide	Yield
Arecanut	1997	Whole Year	Assam	73814.0	56708	2051.4	7024878.38	22882.34	0.796086957
Arhar/Tur	1997	Kharif	Assam	6637.0	4685	2051.4	631643.29	2057.47	0.710434783
Castor seed	1997	Kharif	Assam	796.0	22	2051.4	75755.32	246.76	0.238333333
Coconut	1997	Whole Year	Assam	19656.0	126905000	2051.4	1870661.52	6093.36	5238.051739
Cotton(lint)	1997	Kharif	Assam	1739.0	794	2051.4	165500.63	539.09	0.420909091

Figure 1.1: This is the data head

The table below presents a sample of the dataset used in this analysis. This subset has been included here to provide a representative view of the data's structure and the types of variables analyzed. For clarity and brevity, only a selection of rows and columns is shown. The complete dataset, covering all variables across multiple years, seasons, states, and crop types, is attached in full as an appendix to this report. This larger dataset has been used in all analyses discussed in subsequent sections.

Chapter 2

Exploratory Data Analysis

In this section, we will explore the relationships between key variables in the dataset through a series of visualizations. By employing various plots, we aim to uncover patterns and trends that may exist within the agricultural data, shedding light on how factors such as rainfall, fertilizer usage, crop area, and production levels interact with one another. Visualizing these relationships is crucial for understanding the underlying dynamics of agricultural practices, as well as their impacts on crop yield and overall productivity. The following plots will illustrate these interactions, providing valuable insights that can inform decision-making and policy formulation in the agricultural sector.

2.1 Rainfall and Yield

The relationship between rainfall and crop yield is a fundamental aspect of agricultural productivity, as adequate rainfall is crucial for optimal plant growth and development. Generally, an increase in rainfall is expected to enhance yield by providing necessary moisture for crops; however, this relationship can be influenced by other factors such as soil type, crop variety, and agricultural practices. In the scatterplot below, we visualize the correlation between total rainfall and yield across the dataset. Each point represents a data entry, illustrating how variations in rainfall levels correspond to changes in yield. By examining this plot, we aim to identify potential trends and outliers, offering insights into how rainfall impacts agricultural output.

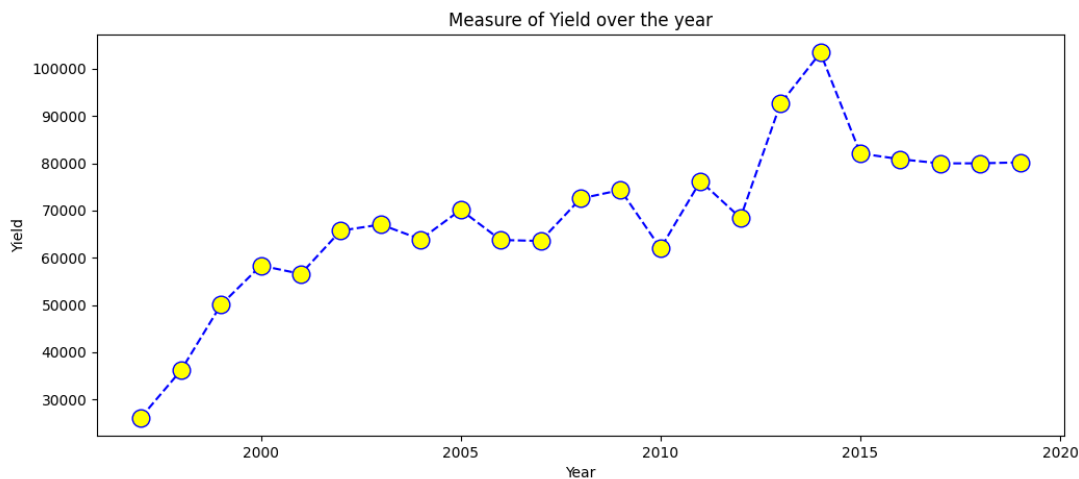


Figure 2.1: Rainfall vs Yield

2.2 Area Under Cultivation Over the Years

The area of cultivation is a critical factor influencing agricultural production, as it directly determines the scale of crop output. Larger cultivated areas typically enable farmers to grow more

crops, assuming optimal conditions such as adequate rainfall and effective agricultural practices. Additionally, expanding the area under cultivation can enhance economies of scale, allowing for better resource allocation, improved machinery usage, and increased labor efficiency. However, it's essential to consider that merely increasing the cultivated area does not guarantee higher production; factors such as soil quality, crop management techniques, and environmental conditions also play significant roles in determining overall yield. Thus, a balanced approach that combines area expansion with sustainable farming practices is crucial for maximizing agricultural production.

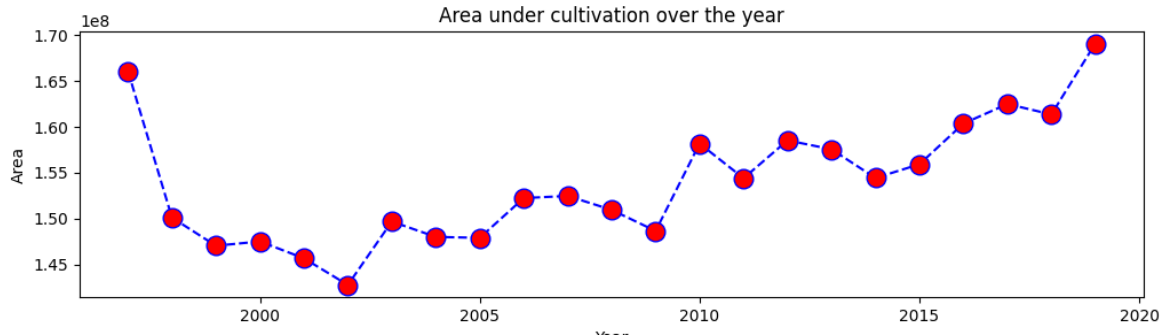


Figure 2.2: Area Under Cultivation Over the Years

2.3 Use of Fertilizers Over the Years

Fertilizers play a vital role in modern agriculture by enhancing soil fertility and providing essential nutrients required for optimal crop growth. They significantly contribute to increased agricultural productivity by supplying macronutrients such as nitrogen, phosphorus, and potassium, which are crucial for plant development. As the global demand for food continues to rise, the use of fertilizers has become increasingly important to ensure that crops can meet this demand while maintaining soil health. In the accompanying plot, we measure the use of fertilizers over the years, which reveals a notable increasing trend. This upward trajectory indicates a growing reliance on fertilizers within the agricultural sector, reflecting both advancements in farming practices and the necessity to boost crop yields to sustain the expanding population. The data underscores the importance of fertilizer application as a key factor in enhancing agricultural productivity and food security.

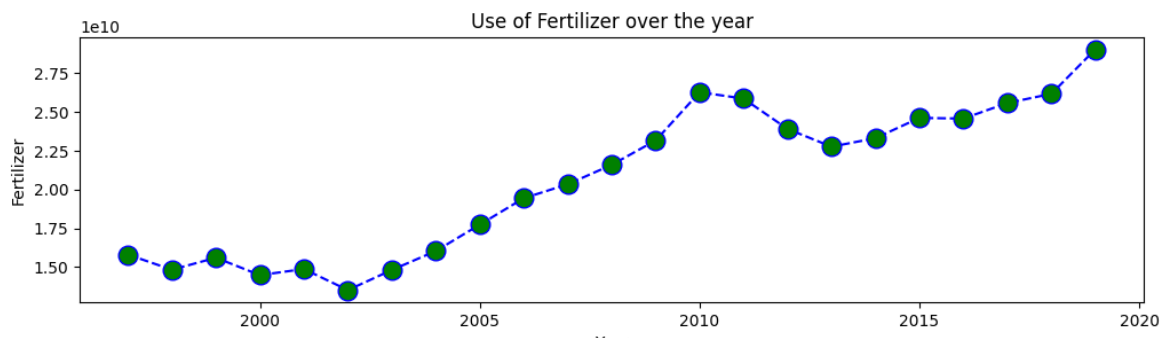


Figure 2.3: Use of Fertilizers Over the Years

2.4 Use of Pesticide Over the Years

Pesticides play a vital role in agriculture by protecting crops from pests, diseases, and weeds that can negatively affect yield and quality. By effectively managing these threats, pesticides contribute to increased crop survival rates and overall productivity, allowing farmers to cultivate a wider variety of crops and maintain higher standards of quality. In the accompanying plot, we analyze the trends in pesticide usage over the years, revealing fluctuations that reflect changing

agricultural practices, regulatory measures, and market demands. This variability underscores the complexities of pesticide application in agriculture, where the emphasis may shift toward integrated pest management strategies and sustainable practices in response to environmental concerns and evolving agricultural challenges.

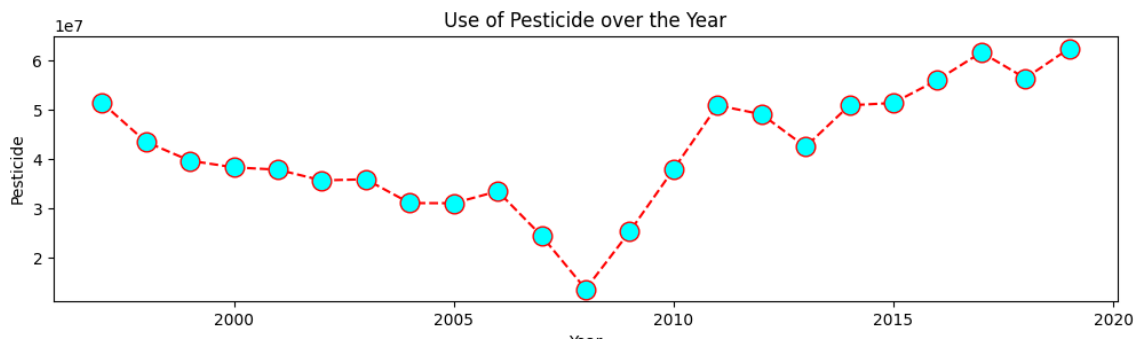


Figure 2.4: Use of Pesticide Over the Years

2.5 State Wise Analysis

The following section presents state-wise plots, offering a detailed look at how various agricultural variables differ across regions. These visualizations allow us to compare trends and patterns in key factors like production, rainfall, fertilizer, and pesticide use across different states. By examining these state-level differences, we gain a better understanding of the regional dynamics in agriculture, which can inform more tailored strategies for improving productivity and sustainability in each area.

2.5.1 Yield in Different States

The following plot illustrates the state-wise crop yields, providing a comparative view of agricultural productivity across different regions. From the plot, we can observe that West Bengal has the highest crop yield among the states. This may be attributed to a variety of factors, such as favorable climatic conditions, soil fertility, efficient water management, and intensive agricultural practices. However, there are likely many contributing factors, and further analysis would be needed to fully understand the underlying reasons for West Bengal's high yield relative to other states.

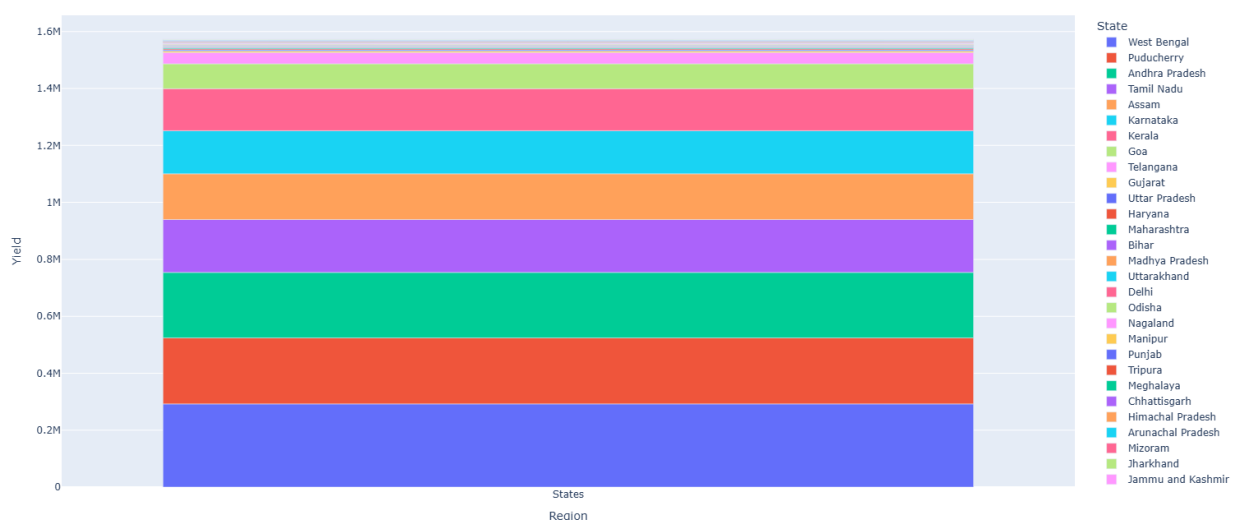


Figure 2.5: Yield in Different States

2.5.2 Annual Rainfall in Different States

The following section includes two plots focusing on annual rainfall patterns across different states. The first plot is dedicated solely to illustrating rainfall levels, highlighting the significant variation in precipitation between regions. The second plot provides a comparative view of rainfall levels alongside crop yield, allowing us to examine potential correlations between rainfall and agricultural productivity. Together, these visualizations offer insights into how regional rainfall influences yield outcomes, which is crucial for understanding and planning sustainable agricultural practices.

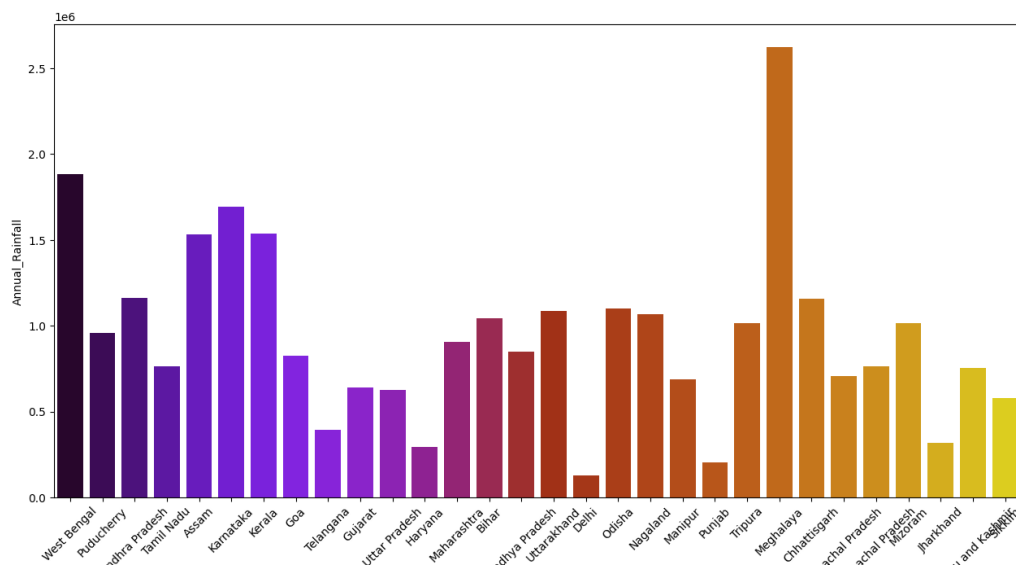


Figure 2.6: State Wise Annual Rainfall

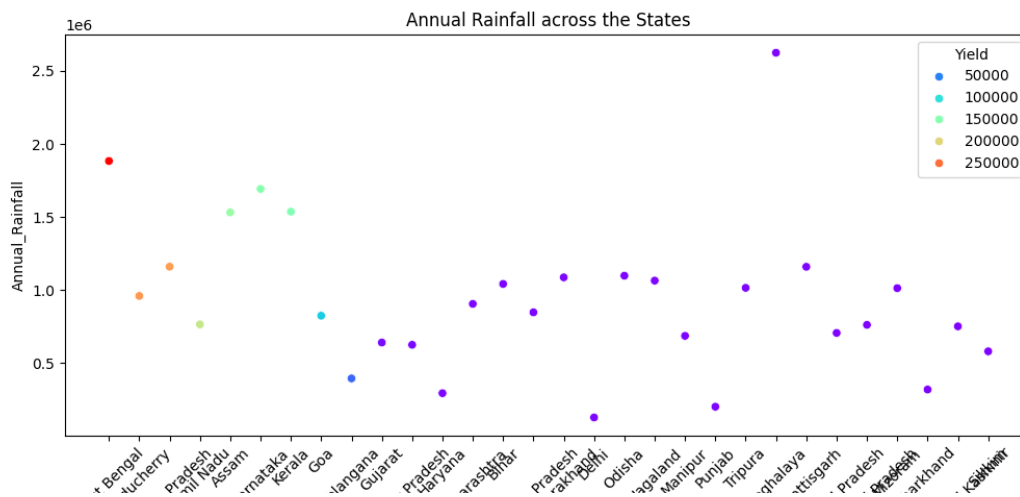


Figure 2.7: Rainfall and Yield

2.5.3 Use of Fertilizers in Different States

The scatterplot below presents a comparison of fertilizer usage and crop yield across different states, allowing us to explore potential correlations between these two variables. Each point represents a state, with its position indicating the level of fertilizer application relative to crop yield. This visualization helps us understand how fertilizer use might contribute to yield variations among states, offering insights into how optimized fertilizer practices could enhance productivity in certain regions.

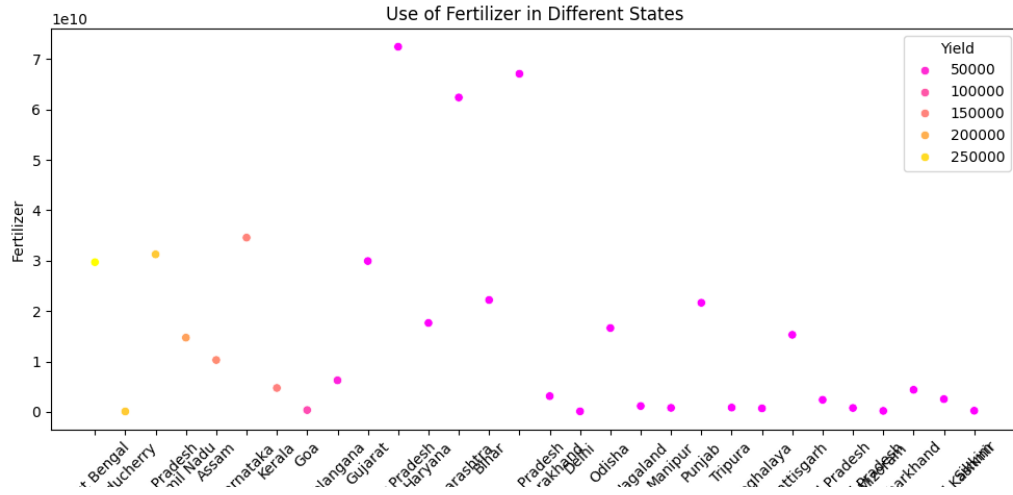


Figure 2.8: Use of Fertilizers in Different States

2.6 Season Wise Analysis

The following section presents season-wise plots, offering a detailed view of how agricultural variables fluctuate across different growing seasons. These visualizations allow us to examine trends and patterns in factors like crop yield, fertilizer use, and rainfall for each season, highlighting the unique challenges and opportunities associated with each period. By analyzing these seasonal variations, we gain valuable insights into how timing and seasonality influence agricultural productivity and resource needs.

2.6.1 Area Under Cultivation in Different Seasons

The barplot below illustrates the area under cultivation across different agricultural seasons, providing a comparative view of how land use varies throughout the year. Notably, we observe that the Kharif season has the maximum area under cultivation, likely due to the reliance on monsoon rains, which support the growth of water-intensive crops like rice. This seasonal variation in cultivated area reflects how climatic factors and crop requirements influence agricultural practices throughout the year.

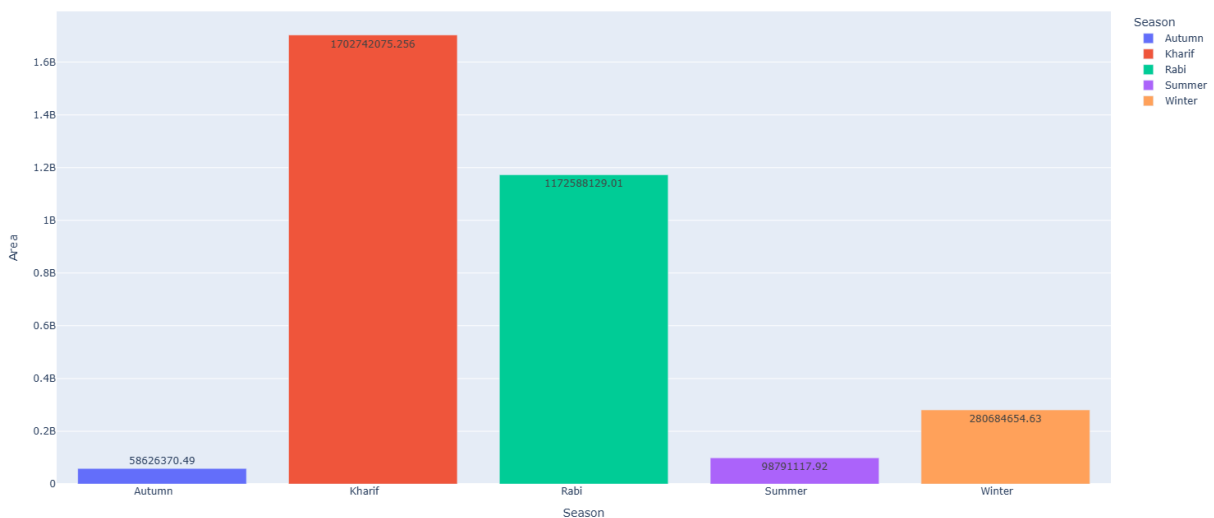


Figure 2.9: Area Under Cultivation in Different Seasons

2.6.2 Yield in Different Seasons

The following pie chart shows the distribution of crop yield across different agricultural seasons, highlighting the contribution of each season to the total annual yield. This visualization allows us to see how productivity varies with seasonal conditions, as well as the relative importance of each growing season in meeting overall agricultural output. By examining the seasonal yield distribution, we can gain insights into the periods of peak productivity and the potential need for targeted support during lower-yield seasons.

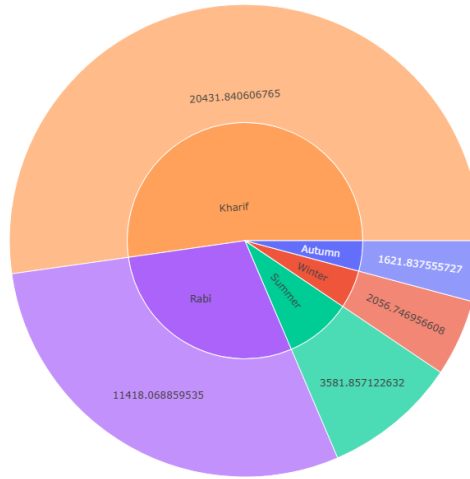


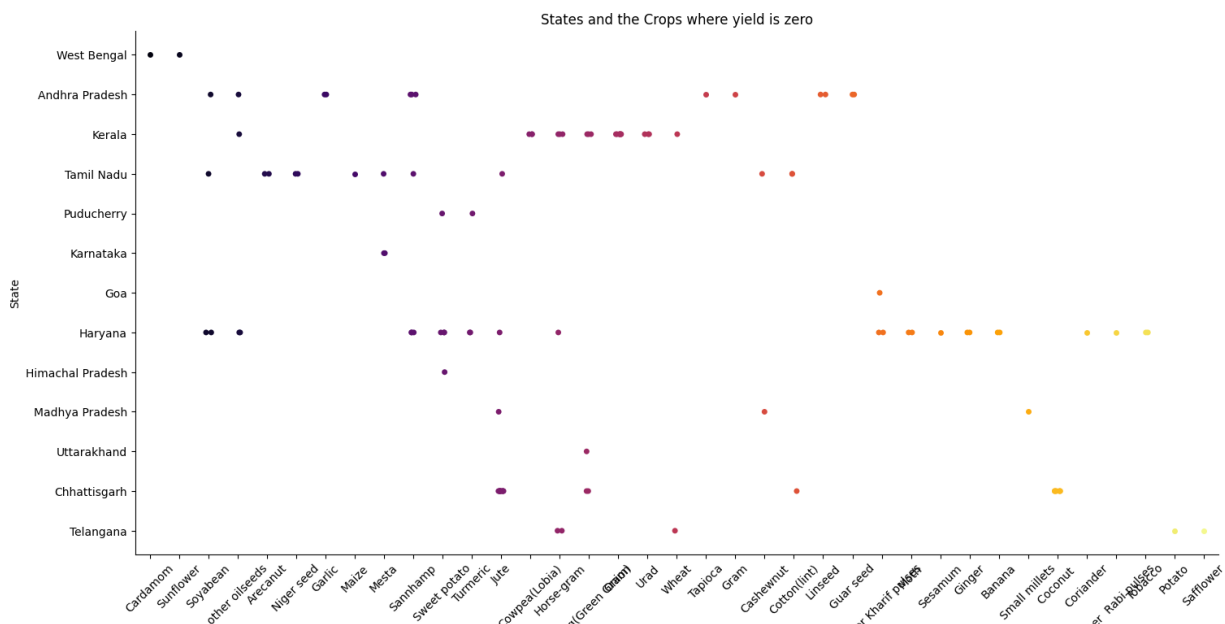
Figure 2.10: Yield in Different Seasons

2.7 Crop Wise Analysis

The next section presents crop-wise plots, offering a focused analysis of various agricultural variables for specific crops. These visualizations allow us to compare factors like yield, area under cultivation, and resource use (such as fertilizer and pesticide application) for each crop. By examining these crop-wise trends, we can gain insights into the unique requirements and productivity patterns of different crops, helping to identify opportunities for optimization and tailored agricultural practices.

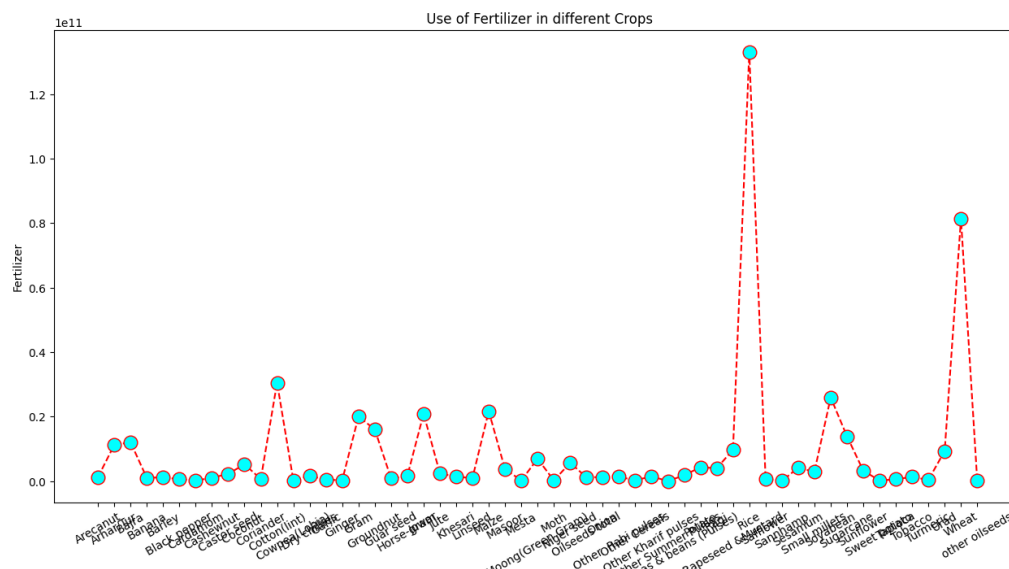
2.7.1 Crops and States where the Yield is Zero

The following plot highlights instances where crop yield is recorded as zero across different states. This visualization helps to identify areas and crops where productivity is critically low or nonexistent, which may be due to factors such as unsuitable climate, soil conditions, pest impact, or resource limitations. By examining these zero-yield cases, we can better understand regional challenges and potential areas where intervention or support may be necessary to improve agricultural outcomes.



2.7.2 Use of Fertilizer in Different Crops

The upcoming plot illustrates the usage of fertilizers across various crops, providing a comparative view of how fertilizer application differs among different agricultural products. This visualization allows us to see the relationships between specific crops and their respective fertilizer needs, highlighting which crops require higher levels of fertilization to achieve optimal yields. Understanding these patterns is essential for developing effective fertilization strategies and ensuring sustainable agricultural practices that maximize productivity while minimizing environmental impacts.



2.7.3 Area Under Cultivation for Different Crops

The following plot presents the area under cultivation for different crops, offering a comparative analysis of land use across various agricultural products. This visualization highlights the relative extent of land dedicated to each crop, reflecting factors such as market demand, regional suitability, and farmers' preferences. By examining the area under cultivation for different crops, we can gain

insights into agricultural trends, identify dominant crops in specific regions, and understand the implications of land use on overall agricultural productivity and sustainability.

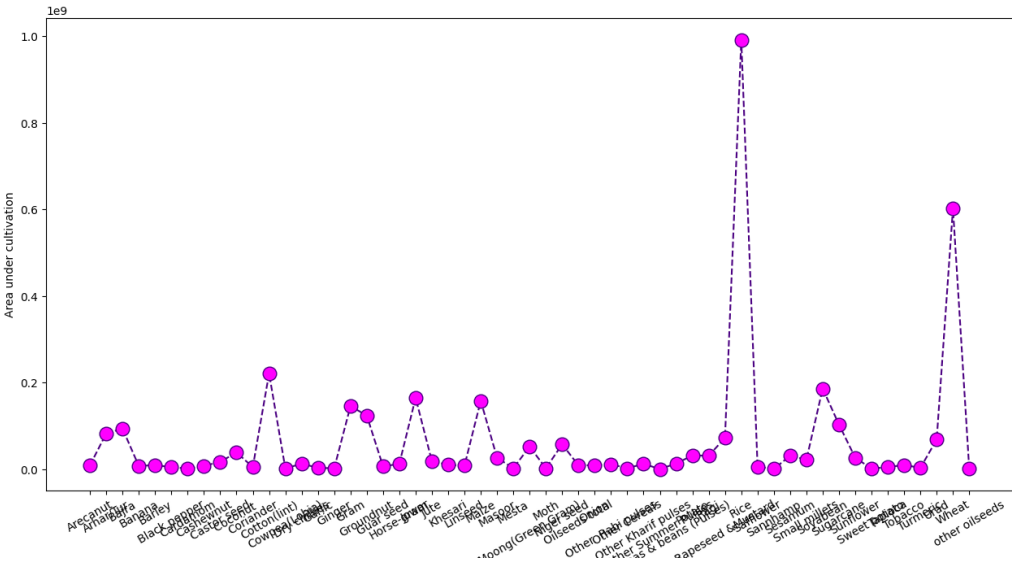


Figure 2.13: Area Under Cultivation for Different Crops

2.7.4 Analysis of Wheat Crop

In this section, we will focus on the wheat crop as a representative example to analyze yield trends over the years. Wheat is one of the most important staple crops globally and plays a crucial role in food security. By examining the historical yield data for wheat, we can identify patterns and fluctuations that may result from various factors such as climatic conditions, agricultural practices, and technological advancements. The following plot illustrates these trends, providing insights into how wheat production has evolved and the challenges faced by farmers in optimizing yield over time.

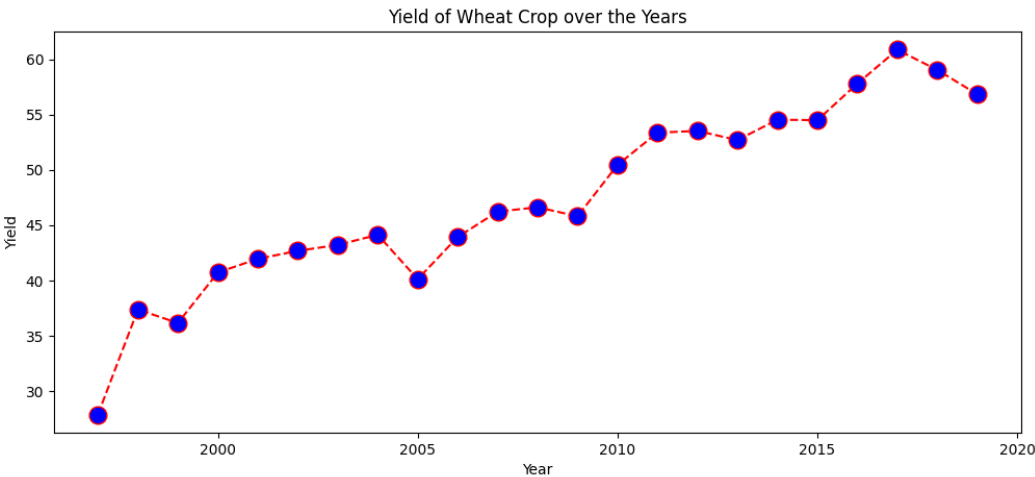


Figure 2.14: Analysis of Wheat Crop

Chapter 3

Predicting Production of Crops

In this analysis, we have chosen multiple linear regression as the predictive modeling technique for estimating crop production due to its ability to accommodate multiple independent variables simultaneously. This approach is particularly advantageous in agricultural contexts, where crop yield is influenced by a variety of factors such as rainfall, area under cultivation, and fertilizer usage. By employing multiple linear regression, we can quantify the individual contributions of each variable to the overall production outcome, allowing for a more nuanced understanding of their interactions. Additionally, this method provides a straightforward interpretation of the results, enabling us to identify significant predictors and assess their impact on crop yields. Given the complexity of agricultural systems, multiple linear regression offers a robust framework for making informed predictions that can guide decision-making and improve agricultural practices.

3.1 Checking for Multicollinearity

To predict crop production effectively using multiple linear regression, it is crucial to first examine the correlation between the variables involved in the analysis. Understanding these relationships helps identify which factors significantly influence crop yield and can guide the selection of predictors for the regression model. Upon analyzing the correlation, I observed a high degree of similarity between the effects of fertilizer and pesticide usage on yield. This redundancy prompted me to drop the pesticide column from the dataset to simplify the model and avoid multicollinearity, ensuring a more robust analysis. Multiple linear regression is particularly useful in this context as it allows us to quantify the relationships between several independent variables and a dependent variable, providing a clearer picture of how each factor contributes to overall crop production. This method enables us to make informed predictions and decisions based on the identified relationships. For a visual representation of the correlations among the variables, I have included a correlation heatmap at the end of this paragraph.

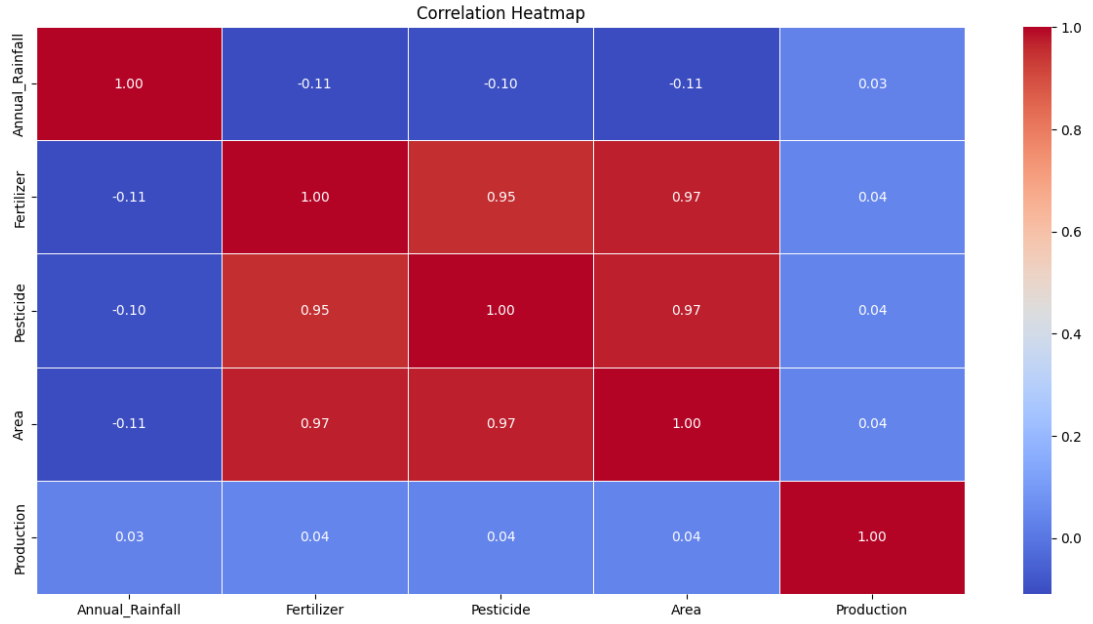


Figure 3.1: Correlation Heatmap

3.2 Checking the Distribution of the Data

The following section presents four empirical histograms illustrating the distributions of the three predictor variables—Annual Rainfall, Area Under Cultivation, and Fertilizer usage—as well as the prediction variable, Crop Production. These histograms provide a visual representation of the empirical distributions, allowing us to observe the underlying patterns and variability within each variable. Analyzing these distributions is essential for understanding the data’s characteristics, identifying any skewness or outliers, and ensuring that the assumptions of multiple linear regression are met. By examining these histograms, we can gain insights into how each predictor variable may contribute to crop production predictions. We can clearly observe from the plot that the data is right skewed.

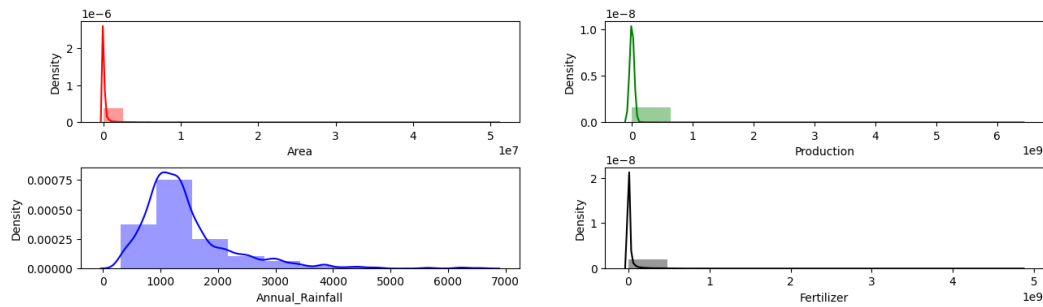


Figure 3.2: Distribution of the Data

3.3 Summary of the Multiple Linear Regression

The purpose of this report is to present the findings from a Multiple Linear Regression (MLR) analysis conducted to understand the relationship between several independent variables (predictors) and a dependent variable (response). This analysis aims to assess how well the predictors can explain variations in the response variable. In the analysis we have specified the state to be West Bengal and the crop to be rice in order to avoid the zero yield entries in the data and make the prediction more accurate. It can be extended via changing for any crop and any state.

3.3.1 Data Description

The dataset used for this analysis contains the following variables:

- **Dependent Variable (Y):**
 - *Production*: The quantity of agricultural product produced (in tons).
- **Independent Variables (X):**
 - *Rainfall*: Amount of rainfall (in mm).
 - *Area*: Area cultivated (in hectares).
 - *Fertilizer*: Amount of fertilizer used (in kg/ha).

3.3.2 Model Development

A Multiple Linear Regression model was developed using the `scikit-learn` and `statsmodels` libraries in Python. The model was fitted using the following equation:

$$\text{Production} = \beta_0 + \beta_1 \cdot \text{Rainfall} + \beta_2 \cdot \text{Area} + \beta_3 \cdot \text{Fertilizer} + \epsilon \quad (3.1)$$

Where:

- β_0 is the intercept,
- $\beta_1, \beta_2, \beta_3$ are the coefficients for the independent variables,
- ϵ is the error term.

3.3.3 Model Fitting

The following Python code was used to fit the model:

```
# Importing Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('crop_yield.csv')
data = df[['Annual_Rainfall', 'Fertilizer', 'Pesticide', 'Area', 'Production']]
# Calculate the correlation matrix
correlation_matrix = data.corr()

df1 = df.copy()
df1 = df1.drop(['Pesticide'], axis = 1)
```

```

df2 = df1[(df1['Crop']=='Rice') & (df1['State']=='West Bengal')]
x = df2[['Area', 'Annual_Rainfall', 'Fertilizer']]
y = df2[['Production']]

#split the data into training and test set
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size = 0.2, random_state = 42)

from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

lr = LinearRegression()
lr.fit(x_train, y_train)

y_pred_train = lr.predict(x_train)
print("Training Accuracy : ", r2_score(y_train, y_pred_train))

y_pred_test = lr.predict(x_test)
print("Test Accuracy : ", r2_score(y_test, y_pred_test))
print("Coefficients:", lr.coef_)
print("Intercept:", lr.intercept_)

```

3.3.4 Model Summary

The following table summarizes the output from the regression analysis:

Statistic	Value
R-squared(test)	0.9663812151764318
Coefficients	
Rainfall	-0.16265163
Area	1.3702517
Fertilizer	0.00777246
Intercept	427171.04181

Table 3.1: Model Summary

3.3.5 Residual Plot

After fitting the multiple linear regression model, it is crucial to evaluate the model's assumptions and performance. One of the key diagnostic tools for this purpose is the residual plot. A residual plot displays the residuals—the differences between observed and predicted values—against the predicted values or an independent variable.

The significance of examining residual plots lies in their ability to reveal patterns that might indicate issues with the model. Ideally, the residuals should be randomly scattered around zero, suggesting that the model's assumptions of linearity, homoscedasticity (constant variance), and independence of errors are satisfied. If any discernible patterns, trends, or clusters are observed in the residual plot, it may suggest potential problems such as non-linearity, heteroscedasticity, or the presence of outliers. This visual assessment helps guide further model refinement and ensures the reliability of predictions derived from the regression analysis.

Now, let's take a look at the residual plot to assess the adequacy of our multiple linear regression model.

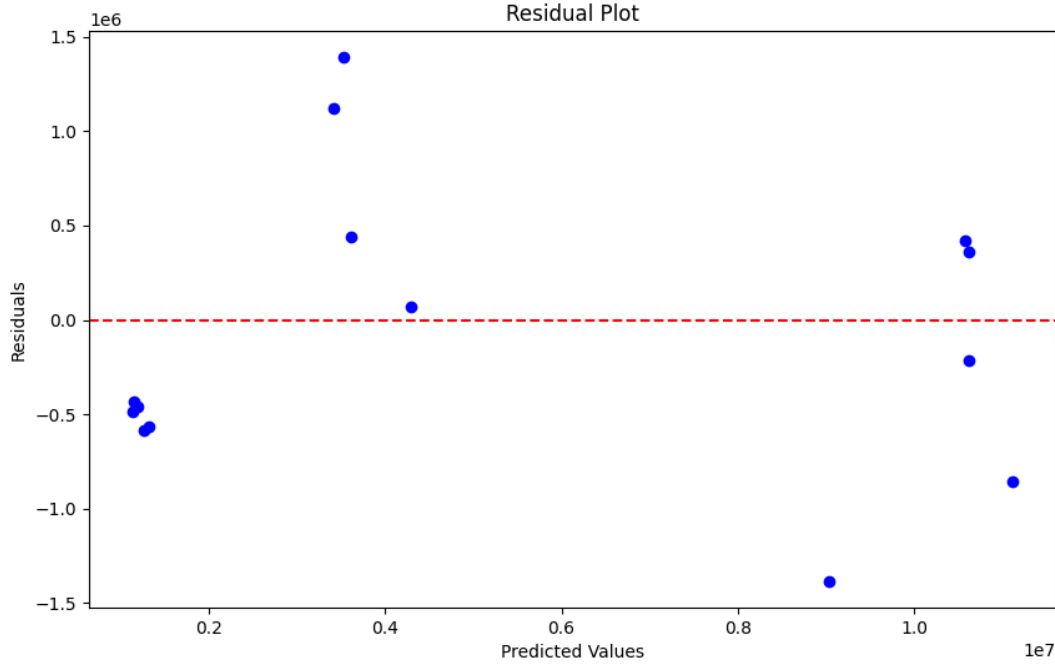


Figure 3.3: Residual plot

3.3.6 Interpretation of Results

- **R-squared:** The R-squared(test) value indicates that 96.63812151764318% of the variability in production is explained by the independent variables.
- **Coefficients:** The coefficients for each variable represent the expected change in production for a one-unit increase in that variable, holding all other variables constant:
 - *Rainfall:* For each additional mm of rainfall, production increases by -0.16265163 (i.e. decreases).
 - *Area:* For each additional hectare of area cultivated, production increases by 1.3702517.
 - *Fertilizer:* For each additional kg of fertilizer used per hectare, production increases by 0.00777246.

3.4 Conclusion

In conclusion, this project has effectively utilized multiple linear regression analysis to examine the relationships between various agricultural factors and crop production. Through careful modeling and analysis, we have gained valuable insights into how predictors such as rainfall, area cultivated, fertilizer usage, and pesticide application influence crop yield across different seasons, states, and crops.

The findings underscore the significance of these variables in enhancing agricultural productivity, providing a foundation for data-driven decision-making for stakeholders in the agricultural sector. Furthermore, the project emphasizes the importance of employing diagnostic tests to validate the model's assumptions and improve the robustness of predictions.

For those interested in replicating or extending this analysis, the detailed codes and dataset used in this project are provided in the attached zip file. Additionally, the original source of the data can be accessed through the following link: <https://www.kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset/data>.

By making this information available, we hope to contribute to ongoing research and discussions surrounding agricultural practices and the effective utilization of data in this vital field.